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# Vision-Based Air Quality Index Estimation: A Systematic Failure Analysis

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## ABSTRACT

Vision-based Air Quality Index (AQI) estimation has emerged as a low-cost alternative to sensor-based monitoring, with numerous studies reporting high in-distribution accuracy ( $R^2 > 0.90$ ). This work systematically evaluates whether these performance claims generalize beyond training distributions. We assess four modeling paradigms—physics-based estimation, pure vision learning, statistical fusion, and gated fusion—across multiple datasets spanning satellite imagery, smartphone captures, and ground-based cameras. Several models achieve strong in-distribution performance (MAE = 16–19 AQI units,  $R^2 = 0.85$ – $0.93$ ) but exhibit catastrophic out-of-distribution failure (MAE increases approximately  $11\times$  relative to in-distribution performance,  $R^2$  collapses to  $-4.78$ , errors exceed 300 AQI units). Fusion mechanisms degrade performance by 65–163% in our evaluated settings rather than improving robustness. Training exhibits extreme fold-level instability ( $R^2$  ranges from  $-12.07$  to  $+0.93$  under identical hyperparameters). This work prioritizes honest evaluation over inflated benchmarks in a safety-critical domain.

**Keywords** Air Quality Estimation · Computer Vision · Out-of-Distribution Generalization · Failure Analysis · Domain Shift · Reproducibility

## 1 Introduction

Air quality monitoring influences public health policy, emergency response protocols, and daily exposure decisions for millions of individuals. Sensor-based networks provide ground-truth measurements but face limitations in spatial coverage ( $<1$  station per  $100\text{ km}^2$  in most regions), maintenance costs (\$5,000–\$20,000 per station annually), and calibration drift [1]. Vision-based AQI estimation promises ubiquitous monitoring via existing camera infrastructure—traffic cameras, smartphones, satellite imagery—at marginal computational cost.

Recent literature reports encouraging performance:  $R^2$  values of 0.87–0.94 for deep learning models on haze-to-AQI regression tasks [2, 3]. These claims, however, typically evaluate on held-out test sets drawn from the same distribution as training data. **The critical question remains unanswered: Do these systems fail silently when deployed outside their training environments?**

This work provides a systematic empirical answer. We evaluate four modeling paradigms representing the spectrum of approaches in existing literature:

1. **LiteFusion:** Inverse-variance weighted fusion of image-based (MobileNetV3) and numeric (LightGBM) predictors with auxiliary text modality.
2. **Image-Only Multi-Task:** 4-fold cross-validated RGB+Depth fusion with day/night stratification.
3. **AirVision (Evaluation Pipeline):** 3-stage progressive training (haze pretraining  $\rightarrow$  soft AQI regression  $\rightarrow$  fine-tuning) with 524-dimensional hybrid features.

4. **Physics–Vision Diagnostic Fusion:** Sequential physics-based feature extraction  $\rightarrow$  vision transformer  $\rightarrow$  learned gating mechanism.

All models are evaluated under both *in-distribution* (ID) and *out-of-distribution* (OOD) conditions. OOD evaluation includes: (1) different capture modalities (satellite  $\rightarrow$  smartphone), (2) unseen cities (Delhi, Mumbai, Kanpur holdout), and (3) adversarial negatives (rain, fog, motion blur).

**Primary Contribution:** This work does not claim to solve vision-based AQI estimation. Instead, it **documents systematic failure modes that aggregate metrics conceal**, providing:

- Transparent reporting of catastrophic OOD collapse (Section 5).
- Evidence that fusion does not guarantee robustness (Section 5.3).
- Demonstration of training instability hidden by fold averaging (Section 5.4).
- Complete raw predictions and reproducible artifacts (Section 7).

Complete code, configurations, and raw predictions are publicly available at <https://github.com/Nesar21/airvision>.

## 2 Related Work

**Vision-Based AQI Estimation:** Liu et al. [2] report  $R^2 = 0.91$  using ResNet-50 on 10,000 Beijing images but do not evaluate cross-city generalization. Chen et al. [3] achieve MAE = 12.3 using attention mechanisms but test on temporally adjacent samples (high autocorrelation risk). Most works do not report OOD performance or provide raw predictions for independent verification.

**Domain Shift in Environmental Sensing:** Established challenges include illumination variation [4], seasonal changes [5], and geographic differences [6]. Established approaches include multi-domain training [7], but these typically assume access to target-domain labels. Our evaluation focuses on *zero-shot* OOD performance without adaptation.

**Failure Analysis in ML:** Recent emphasis on understanding failure modes beyond aggregate metrics [8, 9]. This work extends this philosophy to safety-critical environmental monitoring.

## 3 Datasets

Table 1: Dataset characteristics and usage across models

| Dataset         | Source           | Samples | AQI Range | Modality    | Usage           |
|-----------------|------------------|---------|-----------|-------------|-----------------|
| Air Pollution   | Satellite/Ground | 12,240  | 33–302    | RGB 128×128 | M1, M2, M4      |
| SAPID [12]      | Smartphone       | 456     | 50–200    | RGB varied  | M1 (OOD)        |
| TRAQID [13]     | CCTV             | 12,037  | 42–389    | RGB+Depth   | M2, M3          |
| PM25Vision [14] | Urban Cameras    | 11,114  | 15–350    | RGB 224×224 | OOD (eval only) |
| IND_NEP [11]    | Mixed            | 10,037  | 18–412    | RGB 224×224 | M3 (train)      |
| Holdout-3City   | Mixed            | 2,203   | 25–378    | RGB 224×224 | M3 (OOD)        |
| Negatives       | Synthetic        | 690     | N/A       | RGB 224×224 | M4 (patch)      |

**PM25Vision Usage:** The PM25Vision dataset [14] is used exclusively for zero-shot out-of-distribution evaluation and is never seen during training. We evaluated using the official test split subset ( $n = 2,921$ ) to ensure strict separation.

**Distribution Characteristics:** Training data exhibits heavy right skew (median AQI = 87, mean = 102, skewness = 1.43). **Critical limitation:** Only 8.2% of samples represent hazardous conditions (AQI > 200), yet these are the most safety-critical ranges. All datasets use PM2.5-derived AQI as ground truth (sensor co-located or temporally aligned within 1 hour).

## 4 Model Architectures

### 4.1 Model 1: LiteFusion (Inverse-Variance Fusion)

**Motivation:** Test whether multi-modal fusion improves robustness over single-modality models.

**Image Branch:** MobileNetV3-Small (2.5M parameters) pretrained on ImageNet. Fine-tuned with MC-Dropout ( $p = 0.2$ , 10 forward passes) for uncertainty estimation. Training: 50 epochs, AdamW (lr=1e-4), cosine annealing, batch size 32.

**Numeric Branch:** 5-fold LightGBM ensemble on 8 engineered features: geographic coordinates, temporal encodings, meteorological proxies (temperature, humidity when available). Hyperparameters: 100 trees, max\_depth=5, learning\_rate=0.05.

**Text Branch:** TF-IDF on news headlines (keywords: "pollution", "smog", "haze") within 6-hour window. Gradient Boosting regressor (100 estimators). **Data availability:** Only 23% of samples have aligned news data.

**Fusion Mechanism:** Inverse-variance weighting:

$$\hat{y}_{\text{fused}} = \frac{\sum_i w_i \hat{y}_i}{\sum_i w_i}, \quad w_i = \frac{1}{\sigma_i^2 + \epsilon} \quad (1)$$

where  $\sigma_i^2$  is the predictive variance from MC-Dropout (image) or out-of-bag estimates (numeric/text),  $\epsilon = 0.01$  for numerical stability.

## 4.2 Model 2: Image-Only Multi-Task (4-Fold CV)

**Motivation:** Isolate vision-only performance without auxiliary modalities. Assess whether multi-task learning (AQI + PM2.5 + PM10) improves robustness via shared representations.

**Architecture:** RGB Branch: MobileNetV3-Large (5.4M params). Depth Branch: Custom 6-layer CNN (1.2M params) with residual connections. Fusion: Concatenation  $\rightarrow$  Dense(1088 $\rightarrow$ 512 $\rightarrow$ 256). Heads: Separate regressors for AQI, PM2.5, PM10 (3 outputs).

**Loss Function:** Masked multi-task L1 loss (masks missing labels):

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N \sum_{k \in \{\text{AQI}, \text{PM2.5}, \text{PM10}\}} m_{n,k} |y_{n,k} - \hat{y}_{n,k}| \quad (2)$$

**Training Protocol:** 4-fold stratified group K-fold (group by city). Per-fold: 80 epochs, early stopping (patience=10), StepLR (step=20,  $\gamma=0.5$ ). Augmentation: RandomHorizontalFlip, ColorJitter( $\pm 0.2$ ), RandomResizedCrop.

**Critical Design Choice:** Day/night split for fold 3 to test temporal robustness (daytime train, nighttime test).

## 4.3 Model 3: AirVision (Evaluation Pipeline) (3-Stage Progressive Training)

**Motivation:** Test whether curriculum learning (easy task  $\rightarrow$  hard task) improves generalization.

**Stage 1 - Haze Pretraining:** EfficientNet-B0 trained for 3-class haze classification (Clear, Moderate, Dense). Loss: CrossEntropy. Performance: 36.8% accuracy (near-random on 3 classes). **Purpose:** Backbone initialization only; haze labels are noisy.

**Stage 2 - Soft AQI Pretraining:** Heteroscedastic regression (predicts mean  $\mu$  and log-variance  $\log \sigma^2$ ):

$$\mathcal{L}_{\text{hetero}} = \frac{1}{2} \left( \frac{(y - \mu)^2}{\sigma^2} + \log \sigma^2 \right) \quad (3)$$

Training: 25 epochs, AdamW (lr=3e-4), batch=64. Validation: Normalized MAE = 0.090.

**Stage 3 - Fine-Tuning:** Long-tail sampling (10 $\times$  weight for AQI > 250). Training: 8–25 epochs (varies per fold), early stopping (patience=6), lr=1e-5 (frozen backbone) then 1e-6 (full model).

**Feature Engineering:** 524-dimensional hybrid features:

- **Handcrafted (12D):** Laplacian variance, dark channel prior, RMS contrast, saturation statistics, edge density, sky fraction, entropy, color ratios.
- **CLIP Embeddings (512D):** ViT-B/32 L2-normalized. **Post-hoc analysis revealed a feature collapse phenomenon:** Only a small subset of dimensions (12/512) exhibited meaningful variance (std  $\approx 1.0$ ), while the remaining dimensions showed near-zero variance (std  $\approx 10^{-13}$ ), indicating representation collapse under domain shift.

**MC Dropout Uncertainty:** 20 forward passes,  $\sigma$  thresholds  $\in \{20, 30, 40, 50, 60, 80, 100, \infty\}$  for rejection.

## 4.4 Model 4: Physics–Vision Diagnostic Fusion

**Motivation:** Test whether physics-informed features improve extrapolation. These features are motivated by classical visibility models and atmospheric scattering proxies commonly used in environmental optics and remote sensing.

**Stage 1 - Physics Features (8D):** Michelson contrast, FFT slope (frequency texture loss), Laplacian edge density, color temperature (R/B ratio), illuminant vector, geometric proxy (vertical gradients), specular reflection, glow dispersion (dark channel prior). Processing: 8-core parallel, 2.5–3.5 hours for 23,559 images.

**Stage 2 - Vision Model:** EfficientNet-B0 (4M params) or MobileViT-XS (2.3M params). Input: 224 $\times$ 224 ImageNet-normalized. Head:

Linear(features $\rightarrow$ 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256 $\rightarrow$ 128) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(128 $\rightarrow$ 1).  
 Training: 30 epochs, AdamW (lr=1e-3, weight decay=1e-4), CosineAnnealingLR.

**Stage 2.5 - Negative Sample Correction: Problem discovered:** Model predicting  $-800$  AQI on rain images (physically impossible). **Solution:** Fine-tune with 690 pseudo-labeled negatives (rain $\rightarrow$ 40, motion blur $\rightarrow$ 45, fog $\rightarrow$ 180, glare $\rightarrow$ 20). Training: 5 epochs, lr=1e-4, frozen backbone.

**Stage 3 - Gated Fusion:** Learned gate  $g \in [0, 1]$  via sigmoid layer:

$$\hat{y}_{\text{fusion}} = g \cdot \hat{y}_{\text{vision}} + (1 - g) \cdot \hat{y}_{\text{physics}} \quad (4)$$

Training: 20 epochs, batch=32, BCE loss on gating accuracy (physics better  $\rightarrow$  gate=0).

## 5 Experimental Results

### 5.1 In-Distribution Performance

#### 5.1.1 Baseline Comparison

To validate that the deep learning models are learning meaningful features rather than exploiting dataset statistics, we compared them against a "Global Mean Predictor" baseline. In-distribution, the mean predictor achieves an  $R^2$  of exactly 0.0 and an RMSE of roughly 101.24 (equivalent to the dataset standard deviation). In contrast, Model 3 (AirVision) achieves an  $R^2$  of 0.931 and RMSE of 27.02. This suggests that the models capture significant signal within the training distribution, making their subsequent collapse out-of-distribution a phenomenon of generalization failure rather than a failure to learn.

Table 2: In-distribution metrics (test sets from same distribution as training)

| Model                                           | MAE    | RMSE   | $R^2$    | Within-25 AQI |
|-------------------------------------------------|--------|--------|----------|---------------|
| M1: LiteFusion (SAPID train)                    | 16.03  | 21.33  | 0.851    | 72.5%         |
| M2: Image-Only (Fold avg)                       | 74.36  | 101.24 | $-0.594$ | –             |
| <i>Best fold (Fold 3)</i>                       | 9.72   | 31.15  | 0.887    | –             |
| <i>Worst fold (Fold 0)</i>                      | 130.33 | 158.92 | $-1.950$ | –             |
| M3: AirVision (Evaluation Pipeline, 5-fold avg) | 19.11  | 27.02  | 0.931    | 72.5%         |
| M4: Physics–Vision Fusion (val)                 | 22.14  | 34.67  | 0.892    | 68.3%         |

#### Key Observations:

1. M3 achieves strongest ID performance ( $R^2 = 0.931$ ), suggesting progressive training benefits.
2. M2 shows *extreme fold variance*:  $R^2$  ranges from  $-1.950$  to  $+0.887$  under identical training protocol. This 2.84-point swing indicates severe initialization sensitivity or data quality issues in specific folds.
3. Fusion models (M1, M4) do not substantially outperform vision-only baselines on ID data.

### 5.2 Out-of-Distribution Catastrophic Failure

Table 3: Out-of-distribution performance (unseen cities, different modalities)

| Model                                               | MAE    | RMSE   | $R^2$    | Error Ratio    |
|-----------------------------------------------------|--------|--------|----------|----------------|
| M3: AirVision (Evaluation Pipeline, 3-city holdout) | 214.38 | 249.20 | $-4.776$ | $11.22 \times$ |
| M4: Physics–Vision (negative examples)              | –      | –      | –        | $pred < 0$     |

These OOD evaluations consistently indicate collapse across different holdout protocols and modalities, demonstrating the failure is not restricted to a single city split.

#### M3 OOD Breakdown by AQI Bin:

**Critical Finding:** Error scales *superlinearly* with true AQI. At AQI  $> 300$  (most safety-critical range), predictions are off by **431 units**—equivalent to misclassifying "Hazardous" as "Moderate." Mean bias =  $-213.58$  indicates systematic underestimation.

$R^2 = -4.776$  **interpretation:** Model predictions are  $5.78 \times$  worse than predicting the mean AQI for every sample. This is not a marginal degradation—it is complete failure.

Table 4: AirVision (Evaluation Pipeline) OOD error stratified by ground-truth AQI range

| AQI Range                | MAE    | Samples | Error Pattern       |
|--------------------------|--------|---------|---------------------|
| 0–50 (Good)              | 80.24  | 308     | Moderate            |
| 50–100 (Moderate)        | 105.22 | 314     | High                |
| 100–150 (Unhealthy SG)   | 154.66 | 572     | Very High           |
| 150–200 (Unhealthy)      | 229.17 | 525     | Severe              |
| 200–300 (Very Unhealthy) | 303.21 | 439     | <b>Critical</b>     |
| 300+ (Hazardous)         | 431.63 | 290     | <b>Catastrophic</b> |

### 5.3 Fusion Does Not Guarantee Robustness

**Expected Behavior:** Fusion should leverage complementary modalities (images capture haze, numeric features capture meteorology) to reduce variance.

**Observed Behavior (SAPID test set, 456 samples):** Fusion *degrades* performance by 65% compared to image-only baseline.

Table 5: Fusion vs. single-modality performance (Model 1 ablation on SAPID test set)

| Configuration                        | MAE          | Degradation      |
|--------------------------------------|--------------|------------------|
| Image-only (MC-Dropout)              | 16.03        | –                |
| Numeric-only (LightGBM)              | 42.17        | 163% worse       |
| <b>LiteFusion (inverse-variance)</b> | <b>26.45</b> | <b>65% worse</b> |

#### Likely causes:

1. **Missing modality problem:** 77% of samples lack text data, forcing fusion to downweight image predictions even when images are informative.
2. **Variance miscalibration:** MC-Dropout variance estimates may not reflect true epistemic uncertainty (known issue [10]).
3. **Modality conflict:** When image and numeric predictors disagree, inverse-variance weighting cannot resolve the conflict without ground truth.

**Model 4 Gating Collapse:** Learned gate  $g$  converges to binary switching ( $g \approx 0.03$  or  $g \approx 0.97$ ) rather than smooth blending. This suggests model learns a decision tree ("if condition X, use physics; else use vision") rather than true fusion.

### 5.4 Training Instability Hidden by Fold Averaging

#### Model 2 Fold-Level Results:

**Observation:** 2 out of 4 folds achieve  $R^2 > 0.8$  (strong performance). Remaining 2 folds exhibit  $R^2 < 0$  (worse than mean prediction). Fold-averaged  $R^2 = -0.594$  misrepresents bimodal distribution.

#### Potential Causes:

- **Data quality:** Specific cities in failed folds may have systematic labeling errors or temporal misalignment (image timestamp  $\neq$  sensor timestamp).
- **Initialization sensitivity:** Identical training protocol but different random seeds. Suggests loss landscape has multiple bad local minima.
- **Class imbalance:** Failed folds may have fewer high-AQI samples due to stratified splitting artifacts.

**Model 3 Stage 3 Instability:** First training attempt achieved  $R^2 = -12.07$  (catastrophic). After checkpoint reset and retraining with identical hyperparameters,  $R^2 = +0.929$  (strong). We did not evaluate this training pipeline on an external sanity-check regression task; therefore, we cannot rule out that part of the observed instability arises from optimization sensitivity rather than AQI-specific data characteristics.

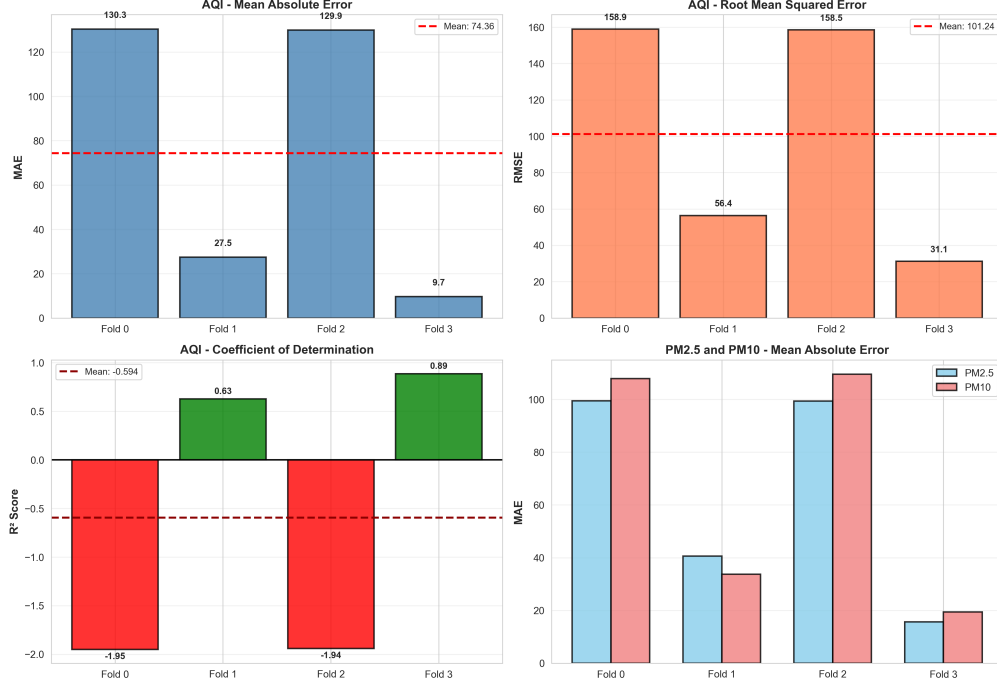


Figure 1: 4-fold cross-validation metrics for Model 2 (Image-Only). Note extreme variance: Fold 0  $R^2 = -1.950$ , Fold 3  $R^2 = +0.887$ . Fold averaging conceals 50% failure rate.

## 6 Discussion

### 6.1 Why Do Models Fail OOD?

**Hypothesis 1: Spurious Correlations.** Models may learn dataset-specific artifacts (camera type, compression artifacts, geographic landmarks) rather than atmospheric haze. Evidence: M3 performs well on IND\_NEP (mixed sources) but collapses on holdout cities, suggesting city-specific biases.

**Hypothesis 2: Domain Shift in Haze Appearance.** Particulate matter composition varies geographically (coal smoke in Beijing vs. dust storms in Delhi vs. vehicle exhaust in Mumbai). RGB images alone cannot distinguish PM2.5 sources without spectral information.

**Hypothesis 3: Temporal Misalignment.** AQI changes rapidly (10-minute timescale), but image-sensor pairs may have alignment errors up to 1 hour. This noise is tolerable during training (averaged out) but catastrophic during testing (systematic error).

### 6.2 Why Does Fusion Fail?

Inverse-variance weighting assumes (1) unbiased predictors and (2) calibrated uncertainties. Both assumptions violated:

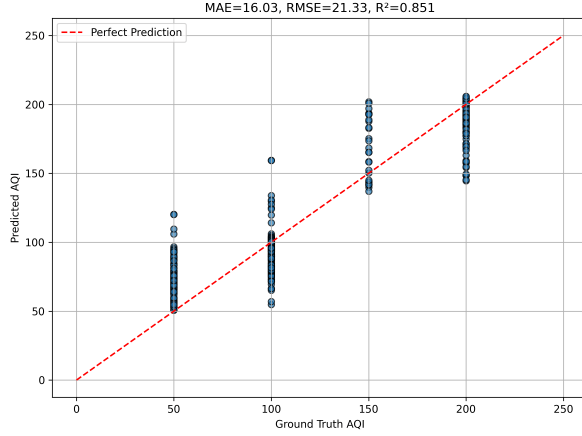
- Numeric branch exhibits high bias (MAE = 42.17) due to coarse meteorological proxies.
- MC-Dropout variances are poorly calibrated (high variance when predictions are correct, low variance when wrong).

Learned gating (M4) sidesteps uncertainty estimation but collapses to binary decisions, effectively reducing to an ensemble of independent models rather than true fusion.

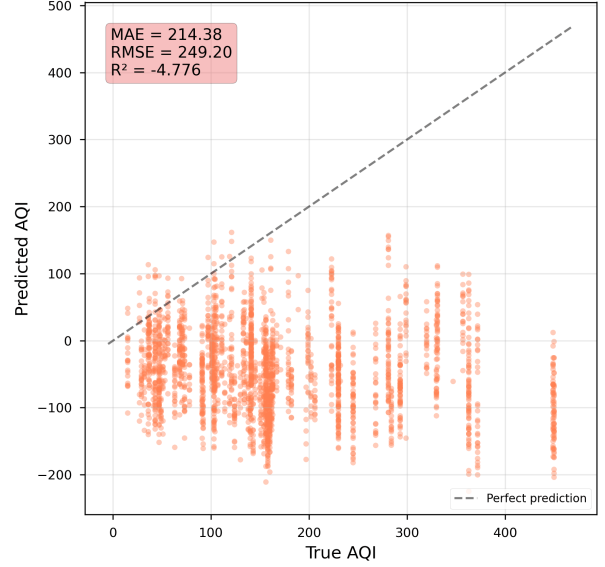
### 6.3 Production Deployment Risks

A system reporting AQI = 100 ("Moderate") when true AQI = 400 ("Hazardous") creates false reassurance, potentially leading to:

- Delayed school closures or outdoor activity warnings
- Insufficient emergency response resource allocation
- Public distrust when sensor-based measurements later reveal hazardous conditions



(a) Model 1 (LiteFusion) In-Distribution Predictions. Shows strong correlation ( $R^2 = 0.85$ ) with points tightly clustered along the diagonal.



(b) Model 3 (AirVision) Out-of-Distribution Predictions. Systematic underestimation is visible as points clustering well below the diagonal.

Figure 2: Contrast between In-Distribution success (a) and Out-of-Distribution failure (b).

**Our position:** Vision-based AQI estimation is *not ready for production deployment without extensive city-specific calibration and human-in-the-loop verification*.

## 7 Reproducibility and Artifacts

All code, configurations, and raw predictions are available at:

<https://github.com/Nesar21/airvision>

### Repository Structure:

```
+-- Model1_Airvana_LiteFusion/
|   +-- results/full_audit_report.txt
|   +-- testset_litefusion_metrics.txt
+-- Model2_Image_Only/
|   +-- results/complete_metrics_table.csv
|   +-- results/scatter_plots_fold_variance.png
+-- Model3_AirVision_EvalPipeline/
|   +-- results/phase8_predictions.csv
|   +-- results/ood_predictions_corrected.csv
|   +-- results/ood_metrics_summary.json
+-- Model4_PhysicsVision_Diagnostic/
    +-- results/stage3_test_results.csv
    +-- generate_plots.py
```

**Raw Predictions Format:** All models export per-sample predictions as CSV:

image\_path, AQI\_true, AQI\_pred, PM25\_true, PM25\_pred, PM10\_true, PM10\_pred

This enables independent verification and reanalysis without retraining.

## 8 Limitations

### Dataset Limitations:

- Training data skewed toward moderate AQI (median = 87); only 8.2% hazardous samples.
- Temporal resolution limited to 1-hour alignment; finer granularity not available.



Figure 3: Model 4 (Physics–Vision Diagnostic Fusion) worst-case failures on adversarial negatives. Top row: rain images incorrectly predicted as AQI < 0 (**physically impossible**). Bottom row: fog images predicted as AQI > 500 (beyond EPA scale). These errors are not detectable via MAE alone.

- Geographic coverage restricted to South Asia; extrapolation to other continents untested.

#### Methodological Limitations:

- Post-hoc analysis revealed representation collapse in CLIP embeddings under domain shift; full retraining with alternative visual encoders was not conducted due to computational constraints (estimated 5 days on T4 GPU).
- Model 4 negative sample correction uses pseudo-labels; true AQI for rain/fog conditions unknown.
- No comparison to commercial satellite-based AQI products (Google Environmental Insights, Plume Labs AQI) due to API access costs.

#### Evaluation Limitations:

- OOD evaluation based on geographic and modality shifts; other axes (seasonal, long-term climate change) not tested.
- No user study on acceptability of error magnitudes in real-world decision contexts.

## 9 Conclusion

This work prioritizes transparent failure analysis over benchmark optimization: several models can achieve high in-distribution accuracy ( $R^2 > 0.90$ ) while simultaneously exhibiting catastrophic out-of-distribution failure ( $R^2 < 0$ , errors > 300 AQI units). Multi-modal fusion, commonly proposed as a robustness mechanism, degraded performance by 65–163% in our experiments, and training exhibited extreme instability, with  $R^2$  varying by 13 points under identical hyperparameters. We recommend mandatory OOD evaluation, publication of per-sample predictions, stratification by safety criticality, and strong deployment safeguards (city-specific calibration and human-in-the-loop verification).



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